DocuFlow Automation / FlowApprove

Enterprise User & Administrator Reference Manual

Version: 2.0.0 (Production Release) | System Platform: Web / Desktop
Engine: FastAPI + Microsoft SQL Server + React Vite
Security Profile: OWASP Top 10 Compliant | Multi-Factor Authentication Active

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Executive Summary & System Architecture

DocuFlow Automation (marketed as FlowApprove) is an enterprise-grade document approval, optical character recognition (OCR), multi-stage workflow automation, and enterprise resource planning (ERP) synchronization platform. Built upon high-throughput Python FastAPI, Microsoft SQL Server, and React with TypeScript, it eliminates manual invoice data entry, automates multi-stage departmental sign-offs, and guarantees financial compliance.

System Architecture & High-Concurrency Pipeline

The platform operates as a cohesive, self-contained architecture with zero external workflow engine dependencies (100% native Python business rules dispatcher).

- **Frontend Layer:** React 18, Vite, TypeScript, Tailwind CSS, Lucide Icons, and React-PDF viewer.
- **Backend Application Server:** Python 3.11 FastAPI running on Uvicorn with AnyIO concurrency limiter tuned to 200 tokens.
- **Database Management System:** Microsoft SQL Server (DocuFlowDB) utilizing SQLAlchemy ORM with a tuned connection pool (pool size 100, max overflow 150).
- **AI & Extraction Engine:** Local/Cloud OCR LLM pipeline reading PDF vectors and raster scans to extract 20+ structured invoice headers and line items.
- **Outbound ERP Sync Hub:** DAPI Sync Hub executing automated HMAC-signed webhooks to third-party ERP platforms (SAP, Oracle, Tally) upon final approval.

> **[SYSTEM PREREQUISITE]** DocuFlow requires modern web browsers (Google Chrome 110+, Microsoft Edge 110+, or Mozilla Firefox 110+). Ensure network access to port 3000 (backend API) and port 5173 / IIS production port.

Chapter 1: Application Access, Login, MFA & Security Management

Purpose: This chapter covers secure access to DocuFlow, Multi-Factor Authentication (MFA), single-session concurrency management, and secure sign-out.

Authorized Users: All registered enterprise users across Clerk, Reviewer, Manager, Controller, and Administrator roles.

How to Access: Navigate your browser to the designated URL (e.g. https://docuflow.internal or http://localhost:5173). If unauthenticated, the system automatically redirects to the Login portal.

Interface Overview & Input Fields

Field / Control	Input Type	Required	Validation Rule	Description
Username / Email	Text input	Yes	Valid user handle or email format	Enterprise login ID assigned by IT.
Password	Password input	Yes	Min 8 chars, hashed via	Secure user password.

			bcrypt	
Remember Session	Checkbox	Optional	Boolean	Persists local authentication claims across browser reloads.
Sign In Button	Button (Primary)	Action	Validates against /api/auth/login	Submits credentials for server verification.
OTP / MFA Token	Numeric / Text	Conditional	6-digit code / time-bounded	Required when Multi-Factor Authentication is active.

Step-by-Step Login & MFA Procedure

1. Open the application URL in a supported web browser.
2. In the Username field, enter your assigned employee ID or enterprise username.
3. Enter your password in the Password field.
4. Click the 'Sign In' button.
5. If MFA is active on your account, an OTP challenge modal appears. Enter the 6-digit code received via your registered email or authenticator app, then click 'Verify OTP'.
6. Upon successful authentication, the system stores your signed JWT bearer token and redirects you to the Executive Dashboard.

Multi-Tab Synchronization & Concurrency Enforcement

DocuFlow integrates an HTML5 BroadcastChannel ('docuflow_auth_channel'). Logging in or out on any single browser tab instantaneously synchronizes session state across all other open tabs without requiring manual page refreshes. Furthermore, single-session concurrency enforcement terminates obsolete sessions if your account is accessed from a separate device.

> **[SECURITY RESTRICTION]** To prevent brute-force attacks, authentication and MFA endpoints enforce a strict rate limit of 15 requests per minute per IP address. Exceeding this limit results in HTTP 429 Too Many Requests.

Chapter 2: Executive Dashboard & Real-Time Analytics

Purpose: Provides real-time visibility into document intake volume, stage bottlenecks, division distribution, and SLA status.

Authorized Users: All authenticated users (data displays scoped according to authorized plant divisions).

How to Access: Click the 'Dashboard' link in the primary navigation header or sidebar.

Metric Card / Chart	Underlying Metric	Filter Behavior	Operational Significance

Total Invoices	Count of active non-deleted documents	Displays total intake volume	Total operational backlog.
Pending Review	Documents at Stage 1 / Intake	Clicks direct to Work Tracker (Pending)	Identifies intake queue depth.
In Approval Flow	Documents actively assigned to approvers	Filters Work Tracker to active workflows	Measures review throughput.
On Hold / Escalated	Documents flagged with exceptions	Filters to status = 'On Hold'	Critical exceptions requiring manager action.
Approved & Cleared	Terminal approved documents	Navigates to Approved Documents archive	Cleared documents ready for ERP export.
Division Distribution	Bar/Pie chart of records per division	Visual breakdown across ACC, VCC, etc.	Identifies departmental volume disparities.

Chapter 3: Document Ingestion & Batch Upload Engine

Purpose: Ingests electronic PDFs and scanned images, executes duplicate SHA-256 validation, and initiates the automated OCR extraction pipeline.

Authorized Users: Clerks, Accounts Payable, Store Keepers, and Administrators.

How to Access: Click the 'Upload' tab on the primary top menu bar.

Control / Field	Type	Required	Description & Constraints
Dropzone Area	Drag & Drop / File Browser	Yes	Accepts .pdf, .png, .jpg, .jpeg, .tiff up to 25 MB per file.
Document Type	Dropdown	Yes	AP INVOICE, PURCHASE ORDER, GOODS RECEIPT NOTE, CUSTOMER FEEDBACK.
Main Division	Dropdown	Yes	Selects parent operating plant (e.g. ACC, VCC, GLOBAL).
Sub-Division / Plant	Dropdown	Optional	Specific operating branch (e.g. ACC_ER, ACC_MDU, ACC_PROD).
Batch Processing	Multi-file queue	Optional	Allows uploading up to 50 documents simultaneously.

Duplicate Prevention Architecture

DocuFlow computes a cryptographic SHA-256 checksum for every uploaded binary file and checks existing database records prior to saving. If an exact file duplicate is detected, the upload is rejected with a message displaying the existing Document ID, preventing redundant financial processing.

Chapter 4: Work Tracker & Active Document Operations

Purpose: The Work Tracker serves as the daily operational clearinghouse where clerks and approvers inspect, verify, and advance active documents.

Authorized Users: Accounts Payable, Reviewers, Approving Managers, and System Admins.

How to Access: Click 'Work Tracker' in the top header or sidebar navigation.

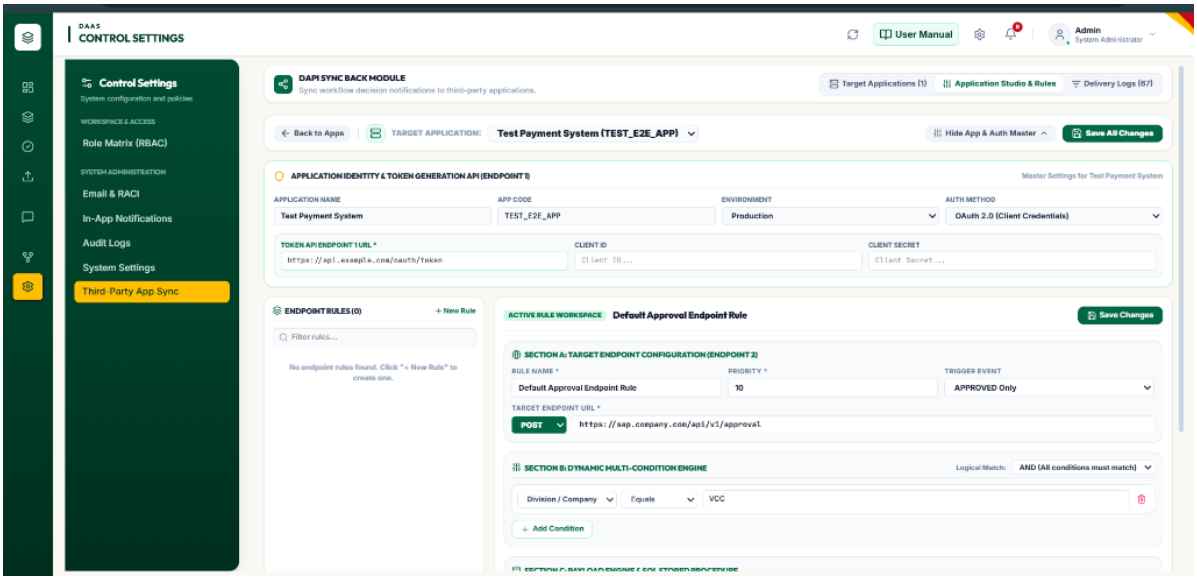


Figure: Work Tracker Table and Document Review Interface

Column Name	Data Source	Filterable	Description
Document ID	Invoice.id / doc_key	Yes (Search)	Unique alphanumeric tracking identifier (e.g. DOC-000172).
Document Type	Invoice.document_type	Yes	Classification type (AP Invoice, PO, GRN, Feedback).
Vendor / Party	Invoice.vendor_name	Yes (Search)	Supplier or business partner name.
Amount	Invoice.total_amount	Yes (Sort)	Net invoice value with formatted currency symbol.
Stage / Step	WorkflowStep.step_name	Yes	Current approval milestone (e.g. Attachment

			Status, Manager Approval).
Current Approver	Invoice.assigned_approver	Yes	User or role pool authorized to take action.
Status	Invoice.status	Yes	Pending, In Progress, On Hold, or Rejected.
Actions	Contextual buttons	No	'Review' opens split-pane editor; 'Download' exports PDF.

Chapter 5: Document Details, OCR Extraction & Verification Form

Purpose: Provides a side-by-side interactive split-pane interface comparing the original document image against OCR-extracted database fields, with full audit trails and checklist enforcement.

Authorized Users: Clerks, Verification Specialists, Approvers, and Auditors.

How to Access: Click 'Review' or double-click any row in the Work Tracker.

Dynamic More Info Field Configuration

DocuFlow features an adaptive More Info configuration engine. Users and administrators can customize which fields appear in the primary top section of the verification view.

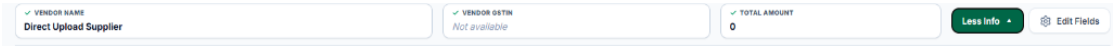


Figure: Dynamic More Info Field Configuration Modal with Category Selection

Field Configuration Options:
• Top 3 Default Fields: Pin key business identifiers (e.g. Purchase Order Number, Payment Terms, GRN Gate ID).
• Scoped Personal Overrides: Standard users can customize their own view without impacting global organization settings.
• Admin Default Promotion: Administrators can set organizational defaults across entire document types.

Available Actions & State Transitions

Button / Action	Target Endpoint	Prerequisites	Database Impact
Approve Stage	POST /api/workflows/approve	Checklist passed & user in approver pool	Advances current_stage +1; if final stage, marks 'Approved' and triggers ERP sync.
Reject Document	POST /api/documents/{id}/reject	Mandatory rejection justification entered	Sets status to 'Rejected', creates audit log, prevents further approvals.

Put On Hold	POST /api/workflows/hold	Mandatory hold justification entered	Sets status to 'On Hold', stops SLA timers, preserves record in Work Tracker.
Send Back / Revert	POST /api/workflows/{id}/send-back	Select target prior stage and comment	Decrements current_stage to designated milestone, alerts previous reviewer.
Reassign Approver	POST /api/workflows/{id}/escalate	Select replacement user/role	Updates assigned_approver in Invoice and approval_assignments tables.

Chapter 6: Approved Documents & Historical Records Archive

Purpose: Provides a permanent, immutable read-only repository of cleared and settled documents with full compliance search filters and PDF export.

Authorized Users: All authenticated users (Clerks, Managers, Financial Auditors).

How to Access: Click 'Approved Documents' in the top header menu.

Search & Filtering Features:
• Date Range Filters: Filter by Year, Month, Specific Date, or Custom From/To date boundaries.
• Division & Type Scoping: Filter specifically by operating plant and document category.
• Full-Text Search: Query by Vendor Name, Invoice Number, or internal tracking key.
• Export to Excel / CSV: Download comprehensive audit-ready spreadsheets containing all financial metadata.

Chapter 7: Specialized Document Types (GRN & Customer Feedback)

Purpose: Details the specialized handling pipelines for Goods Receipt Notes (GRN) and Customer Grievances / Feedback.

Customer Feedback & Grievance Pipeline

The Customer Feedback portal tracks complaints, field inspection reports, and resolution workflows. Unlike standard financial invoices, feedback records require dedicated hold reason taxonomy and root-cause checklists.

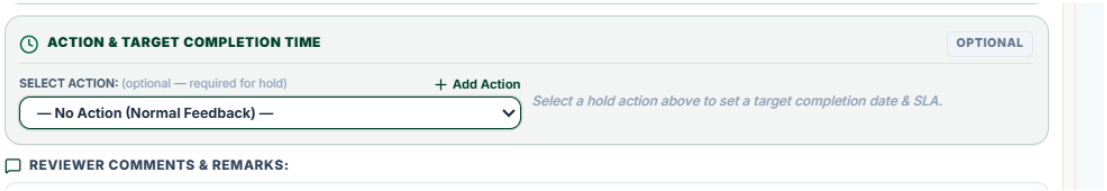


Figure: Customer Feedback Review Dialog with Mandatory Action Selection

> **[COMPLIANCE CONTROL]** To prevent accidental document progression, the Action selector in the Customer Feedback review dialog defaults strictly to empty ('— No Action (Normal Feedback) —'). Submitting a review requires explicit action selection and reviewer justification.

Chapter 8: Flow Builder & Multi-Stage Workflow Canvas

Purpose: Provides visual creation, sequential stage sequencing, approver targeting, and SLA definition for multi-departmental approval paths.

Authorized Users: Workflow Administrators and System Admins (role:manage / workflow:manage permissions).

How to Access: Navigate to 'Control Settings' or 'Workflow Rules' -> 'Flow Builder' tab.

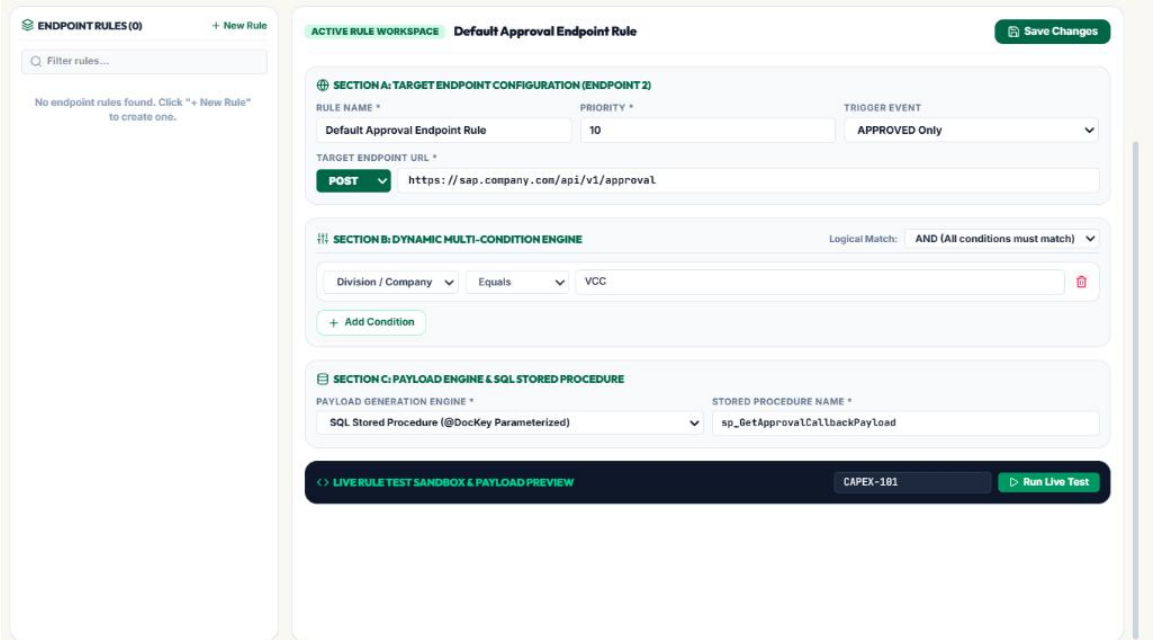


Figure: Flow Builder Multi-Stage Sequential Routing Canvas

Configuration Property	Input Type	Required	Description
Workflow Profile Name	Text string	Yes	Unique workflow identifier (e.g. 'ACC_INVOICE_FLOW', 'CAPEX_HIGH_VALUE').
Stage Number	Integer sequence	Yes	1-indexed order of execution (Stage 1, Stage 2, Stage 3...).
Step / Stage Name	Text label	Yes	Descriptive milestone (e.g. 'Attachment Status', 'Plant Head Approval').

Approver Target	Role / User Handle	Yes	Primary assigned authority (e.g. 'Plant_Finance_Head', 'admin').
Delegate Approver	Role / User Handle	Optional	Fallback approver authorized if primary is on leave or unavailable.
SLA Hours Limit	Numeric integer	Optional	Maximum elapsed time before escalation alert is dispatched.

Chapter 9: Condition Builder & Dynamic Routing Logic

Purpose: Enables financial policy logic (e.g. 'If total amount > 50,000 INR, route to General Manager; otherwise route to Site Accountant').

Authorized Users: Workflow Administrators and Financial Controllers.

How to Access: Navigate to 'Workflow Rules' -> 'Condition Builder' tab.

Supported Operators & Conditions:
• EQUALS (=) / NOT EQUALS (!=): Exact match on division, vendor, or category.
• GREATER THAN (>) / LESS THAN (<): Numeric boundaries on invoice amount or line item quantity.
• CONTAINS: Substring search across vendor names or item descriptions.
• IN LIST: Matches against designated sets of authorized division codes.
• Chaining Logic: Compound rules supported via logical AND and OR operators.

Chapter 10: Checklist Creation & Compliance Templates

Purpose: Enforces physical verification checks (e.g. 'Purchase Order Attached', 'Gate GRN Verified', 'Tax ID Validated') that must pass before stage clearance.

Authorized Users: Compliance Managers, Quality Control, and Administrators.

How to Access: Navigate to 'Workflow Rules' -> 'Checklist Templates' tab.

Checklist Architecture:
• Document-Type Binding: Checklists attach to specific document categories (AP Invoices, GRNs, Feedback).
• Stage-Specific Enforcement: Specific checklist items can be required only at Stage 1 (Intake) or Stage 3 (Final Payment).
• Interactive Pass/Fail Persistence: Toggling checklist items writes directly to the InvoiceChecklistState database table with reviewer attribution.

Chapter 11: Role Creation & Role-Based Access Control (RBAC)

Purpose: Manages enterprise roles, clearance tiers, and granular functional permissions.

Authorized Users: System Administrators.

How to Access: Navigate to 'Control Settings' / 'Admin' -> 'RBAC & Roles' tab.

Permission Token	Functional Area	Capability Granted
document:read	Work Tracker / Archive	Allows viewing document metadata, PDF assets, and audit logs.
document:verify	Document Details Form	Authorizes modifying OCR fields, line items, and checklist states.
workflow:approve	Workflow Actions	Enables clicking 'Approve Stage' to advance documents in assigned stages.
workflow:hold_reject	Exception Actions	Permits placing documents on hold or executing final rejection.
workflow:manage	Flow Builder	Allows creating and publishing multi-stage workflow definitions.
role:manage	Admin RBAC	Grants capability to provision users, edit profiles, and alter clearance matrices.
sync:manage	DAPI Sync Hub	Authorizes configuring ERP webhooks and triggering manual batch syncs.
system:archive	System Maintenance	Enables altering database retention windows and triggering database archival.

Chapter 12: User Management & Scoped Division Provisioning

Purpose: Covers provisioning new enterprise accounts, assigning roles, and scoping user visibility to authorized plant divisions.

Authorized Users: System Administrators and HR / IT Managers.

How to Access: Navigate to 'Admin' -> 'User Management' -> Click '+ User' button.

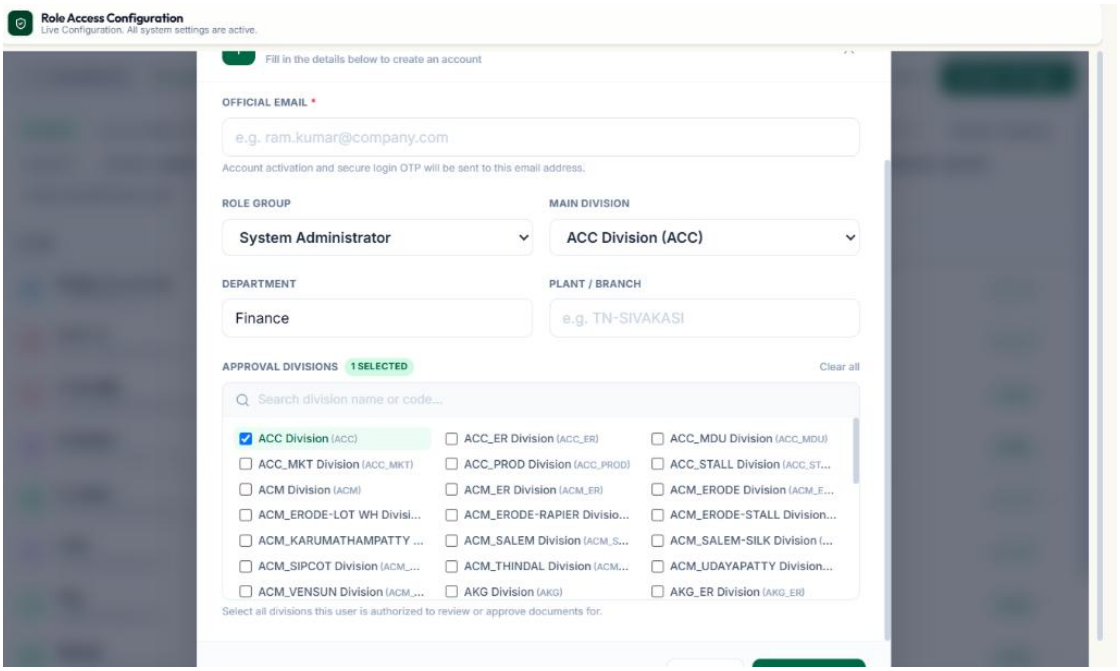


Figure: Admin User Provisioning Interface with Employee Details

Dynamic Approval Division Filtering

To prevent cross-division authorization errors, selecting a 'Main Division' automatically restricts the 'Approval Divisions' multi-select list strictly to operating units belonging to that group.

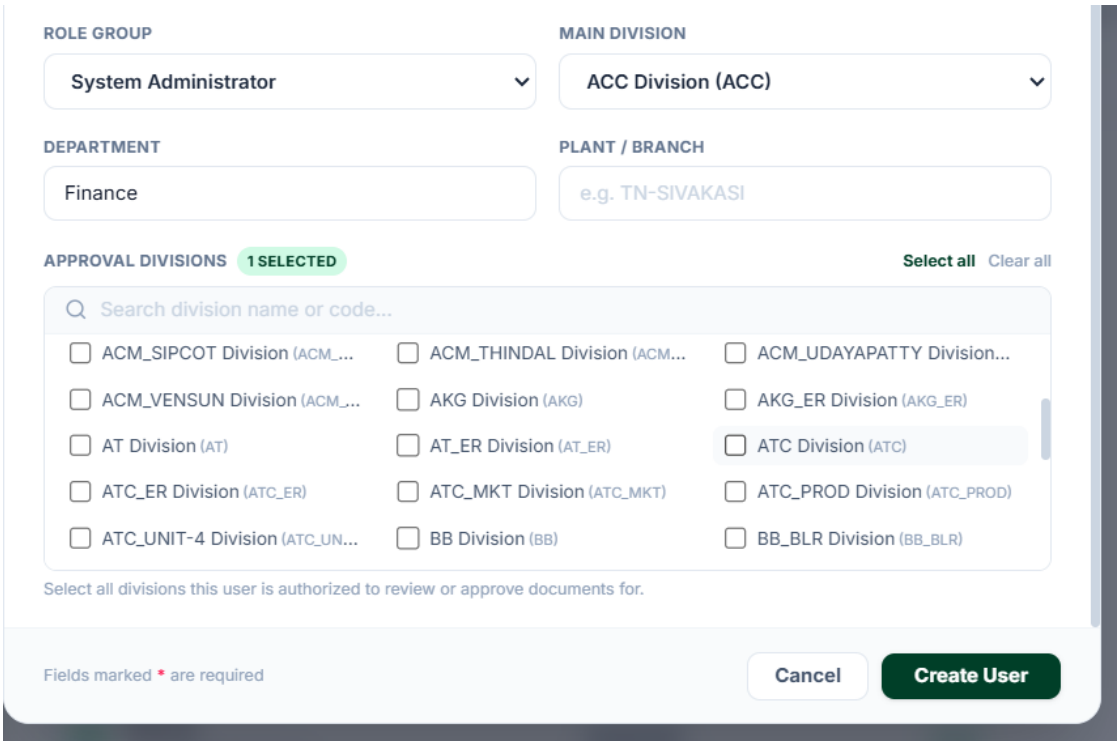


Figure: Main Division Scoping: Selecting ACC Automatically Displays ACC Division Sub-Units

Form Input	Type	Validation Constraints	Behavior & Impact
Employee ID	Text	Unique, non-empty	Primary enterprise employee identifier.
Employee Name	Text	Full legal name	Displays on approval signatures and audit entries.
Email Address	Email	Unique valid email	Destination for MFA tokens and SLA escalation alerts.
Password	Password	Min 8 characters	Securely hashed via bcrypt prior to database storage.
Role	Dropdown	Select from active roles	Assigns clearance permissions (Admin, Manager, Clerk, etc.).
Main Division	Dropdown	Required selection	Determines primary operating group (e.g. ACC, VCC, HQ).
Approval Divisions	Checkbox list	Filtered by Main Division	Grants granular approval authority for specific plant units.

Chapter 13: Third-Party Application Synchronization (DAPI Sync Hub)

Purpose: Manages real-time data exchange, webhook callbacks, and ERP payload synchronization upon final document clearance.

Authorized Users: Integration Engineers, ERP Administrators, and System Admins.

How to Access: Click 'DAPI Sync Hub' / 'Integrations' in the primary navigation header.

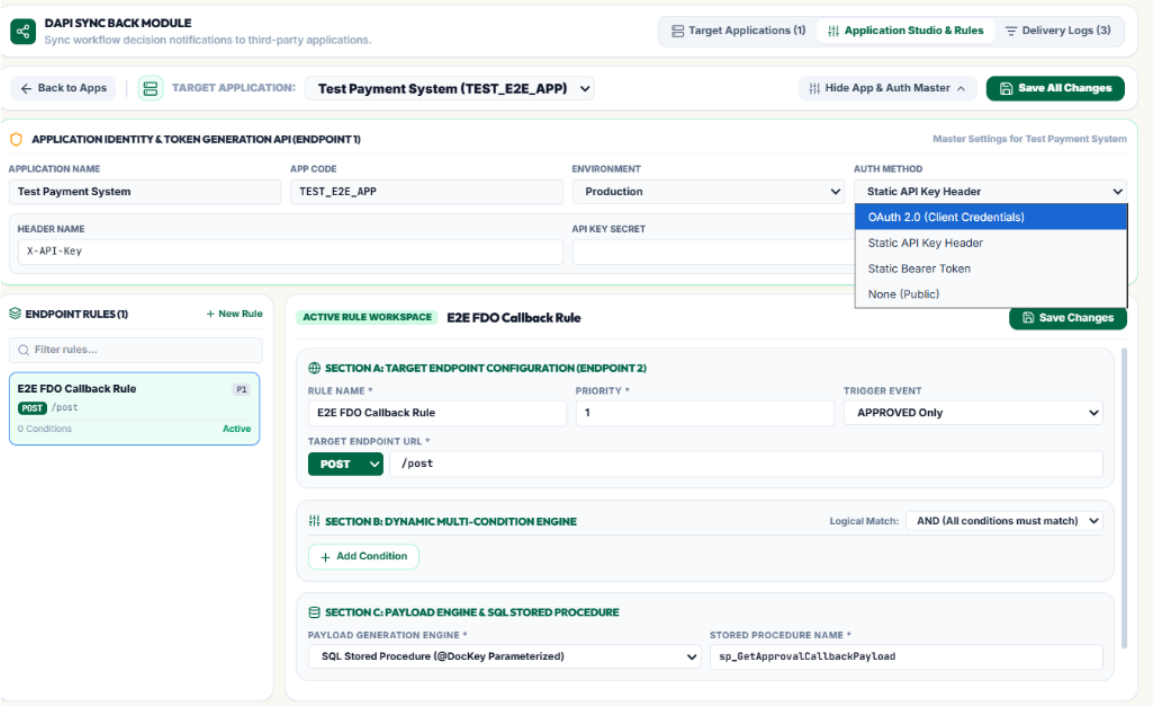


Figure: DAPI Callback Webhook Configuration Interface

Configuration Parameter	Type	Required	Operational Description
Application Name	Text	Yes	Registered consumer (e.g. 'SAP S/4HANA ERP', 'Tally Prime Hub').
Webhook Target URL	HTTPS URL	Yes	External endpoint receiving the approved invoice payload.
Trigger Event	Dropdown	Yes	DOCUMENT_APPROVED, DOCUMENT_REJECTED, or STAGE_COMPLETED.
HMAC Secret Key	Password / Token	Yes	Generates SHA-256 signature in X-Signature header for verification.
Max Retry Count	Integer (1-10)	Yes	Number of automatic retry attempts if target server returns 5xx error.
Retry Interval (s)	Integer (seconds)	Yes	Exponential backoff delay between transmission retries.

Chapter 14: In-App Notifications & RACI Email Alert Matrix

Purpose: Configures automated alert delivery, real-time in-app notification toasts, and RACI matrix escalation routing.

Authorized Users: All users receive alerts; Administrators configure delivery rules.

How to Access: In-App Bell in top header; RACI Matrix in 'Control Settings' -> 'RACI / Notifications'.

Notification Types:
• Document Assigned: Immediate alert when a document enters your approval stage.
• SLA Breach Warning: Notification dispatched when an invoice approaches its configured turnaround deadline.
• Document Rejected / Sent Back: Alerts the uploader or previous reviewer with reason notes.
• ERP Sync Failure: System administrator notification if outbound webhook exhausts maximum retries.

Chapter 15: System Administration, Database Archival & Retention Maintenance

Purpose: Covers automated database archival schedules, audit retention pruning, and system engine diagnostic logs.

Authorized Users: System Administrators and Database Engineers.

How to Access: Navigate to 'Control Settings' -> 'System & Archival' tab.

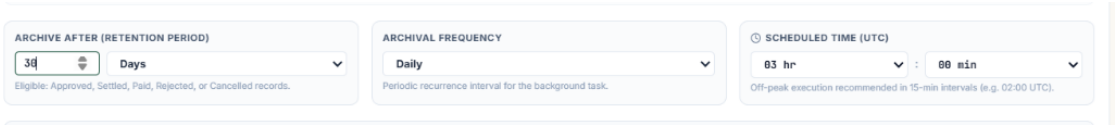


Figure: Automated Database Archival and Retention Scheduler

Setting	Options	Default	Description
Archival Frequency	Daily, Monthly, Yearly	Monthly	Frequency at which background engine migrates settled records.
Scheduled Execution Time	24-hour time (HH:MM)	02:00 AM	Off-peak execution window to minimize database load.
Retention Window	30, 90, 180, 365 Days	90 Days	Documents older than this threshold are packaged into archives.
Archive File Target	File system path / storage URI	/backups/archive/	Destination for compressed archive database files.

Chapter 16: Troubleshooting & Diagnostic Guide

Purpose: Provides actionable remedies for common operational, authorization, and network error codes.

Error Symptom	Probable Root Cause	Step-by-Step Remediation Procedure
HTTP 401 Unauthorized / Token Expired	JWT session expired or invalid credentials	Log out, clear browser cookies, and re-enter credentials with valid MFA token.
HTTP 403 Forbidden on Approval	Current user is not in assigned approver pool	Verify current workflow step definition in Flow Builder; ensure user possesses required role.
Duplicate Document Detected	File with identical SHA-256 hash exists	Search for document key in Work Tracker or Approved archive; do not re-upload identical PDF.
OCR Extraction Field Missing	Low-resolution scan or skewed page	Use rotation buttons in Document Details; click field input box and manually enter correct value.
Outbound ERP Sync Pending	Target third-party server down or invalid HMAC	Open DAPI Sync Hub, inspect transmission log, verify webhook URL, and click 'Retry Sync'.
HTTP 429 Too Many Requests	Exceeded rate limit (15 requests/min)	Wait 60 seconds before retrying login or MFA token verification.

Chapter 17: Comprehensive Glossary & Quick-Reference Sheets

Acronym / Term	Full Definition	Context in DocuFlow
OCR	Optical Character Recognition	Automated text extraction from uploaded PDF and image documents.
LLM	Large Language Model	AI engine parsing unstructured OCR text into structured financial schemas.
RBAC	Role-Based Access Control	Security architecture mapping functional clearances to user roles.
RACI	Responsible, Accountable, Consulted, Informed	Matrix defining stakeholder notification and escalation rules.
DAPI	Document Application Programming Interface	Synchronization hub exchanging data payloads with third-party ERPs.
GRN	Goods Receipt Note	Warehouse physical gate entry verification confirming received merchandise.

SLA	Service Level Agreement	Maximum allowable time window for approvers to clear a document stage.

Chapter 18: Documentation Gaps & Unresolved Evidence

In accordance with strict technical writing standards, the following table explicitly catalogs features that are operational in backend code but have unverified UI visual states or lack direct screenshot evidence in the current release inventory:

System Module	Current Documentation Status	Missing Evidence	Recommended Remediation
Payment Readiness View	Code verified in PaymentReadinessPage.tsx	Direct high-resolution screenshot pending	Capture production screenshot under financial settlement role.
Data Verification Queue	Code verified in DataVerificationPage.tsx	Direct high-resolution screenshot pending	Capture production screenshot under Clerk verification role.
SLA Escalation Alert Toast	Code verified in NotificationEngine.py	Visual capture of active toast alert pending	Trigger artificial SLA timeout during staging demo to snapshot alert.
Active ERP Webhook Ack	Code verified in callback_integrations.py	Production SAP/Oracle 200 OK receipt image pending	Capture network trace screenshot during live staging ERP sync.